

SHWETHA S – PES1UG24CS452– BPS#24

KANBAN PROJECT – JIRA HANDS-ON PRACTICE

Name: SHWETHA S

SRN: PES1UG24CS452

Problem Statement: BPS#24 – Public Bus Live Tracking & Crowding Estimator

Project Name: Kanban_BPS#24

Tool Used: Jira

Project Type: Kanban

1. INTRODUCTION

This document presents the implementation of the **Public Bus Live Tracking & Crowding Estimator** problem statement using a **Kanban project in Jira**.

The project was created using the Kanban template. The requirements were organized using an Epic, Tasks, Stories, and Subtasks. The Jira Kanban board was used to manage and track the work items through different stages such as **To Do, In Progress, In Review, and Done**.

2. PROJECT CREATION

A Kanban project was created in Jira with the following details:

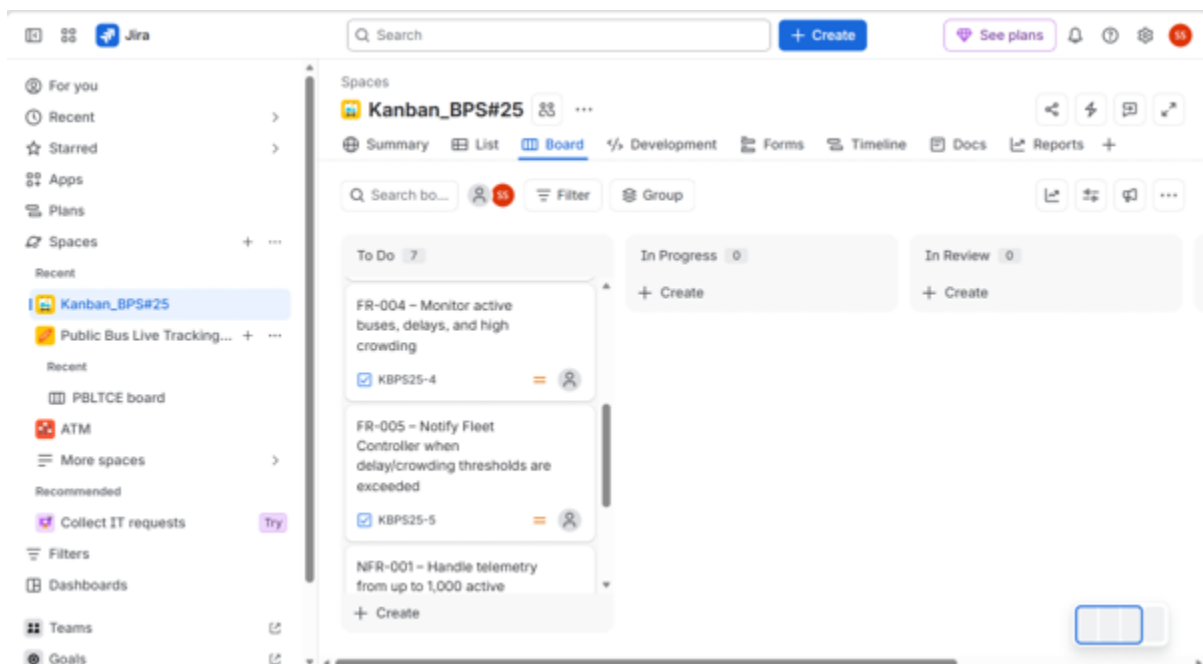
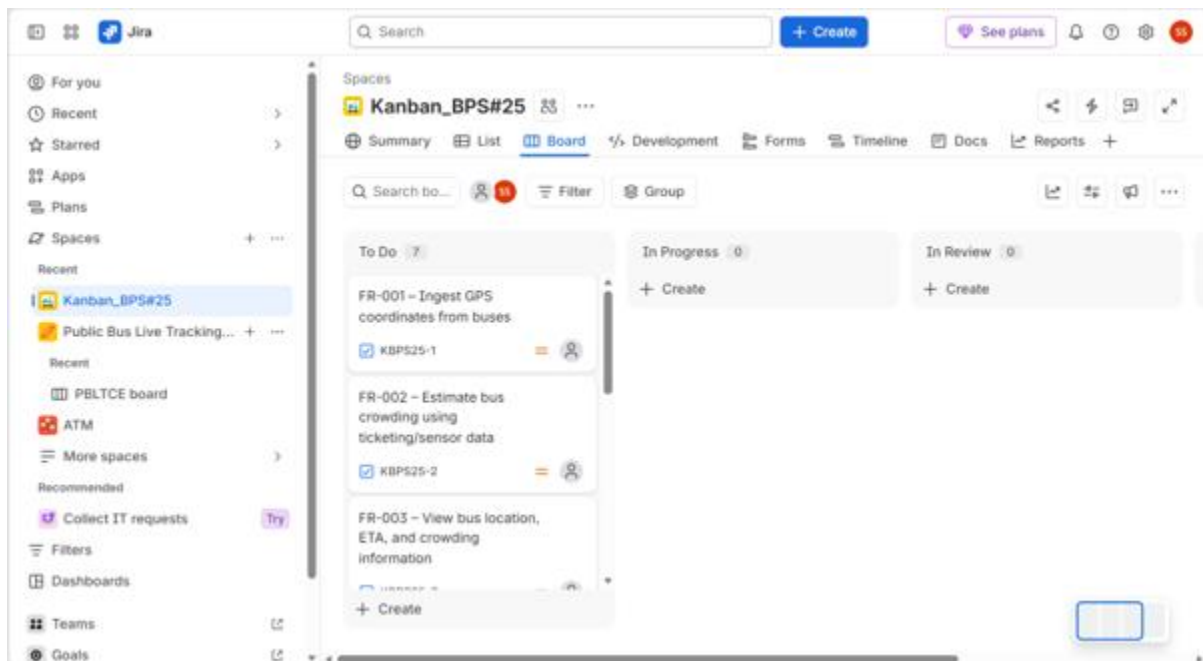
- **Project Name:** Kanban_BPS#25
- **Project Type:** Kanban
- **Problem Statement:** Public Bus Live Tracking & Crowding Estimator
- **Epic:** Public Bus Live Tracking & Crowding Estimator

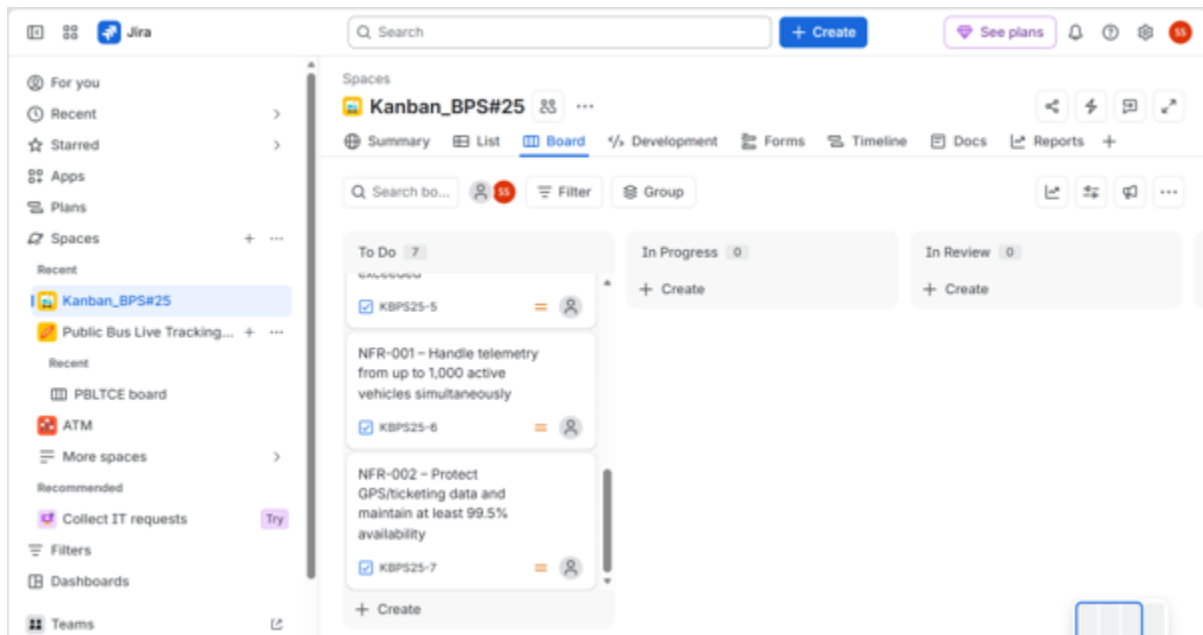
The project board was created successfully and used for managing the requirements and associated work items.

Screenshot 1 – Kanban Board Creation

Description:

The screenshot shows the created Kanban project board in Jira with the required functional and non-functional requirements added to the project.





3. REQUIREMENTS

The following **5 Functional Requirements (FRs)** and **2 Non-Functional Requirements (NFRs)** were created in Jira.

Functional Requirements

1. **FR-001 – Ingest GPS coordinates from buses**
2. **FR-002 – Estimate bus crowding using ticketing/sensor data**
3. **FR-003 – View bus location, ETA, and crowding information**
4. **FR-004 – Monitor active buses, delays, and high crowding**
5. **FR-005 – Notify Fleet Controller when delay/crowding thresholds are exceeded**

Non-Functional Requirements

6. **NFR-001 – Handle telemetry from up to 1,000 active vehicles simultaneously**
7. **NFR-002 – Protect GPS/ticketing data and maintain at least 99.5% availability**

4. EPIC AND REQUIREMENT TASKS

An Epic named **Public Bus Live Tracking & Crowding Estimator** was created.

All seven requirement Tasks were assigned to this Epic. The Tasks represent the functional and non-functional requirements of the system.

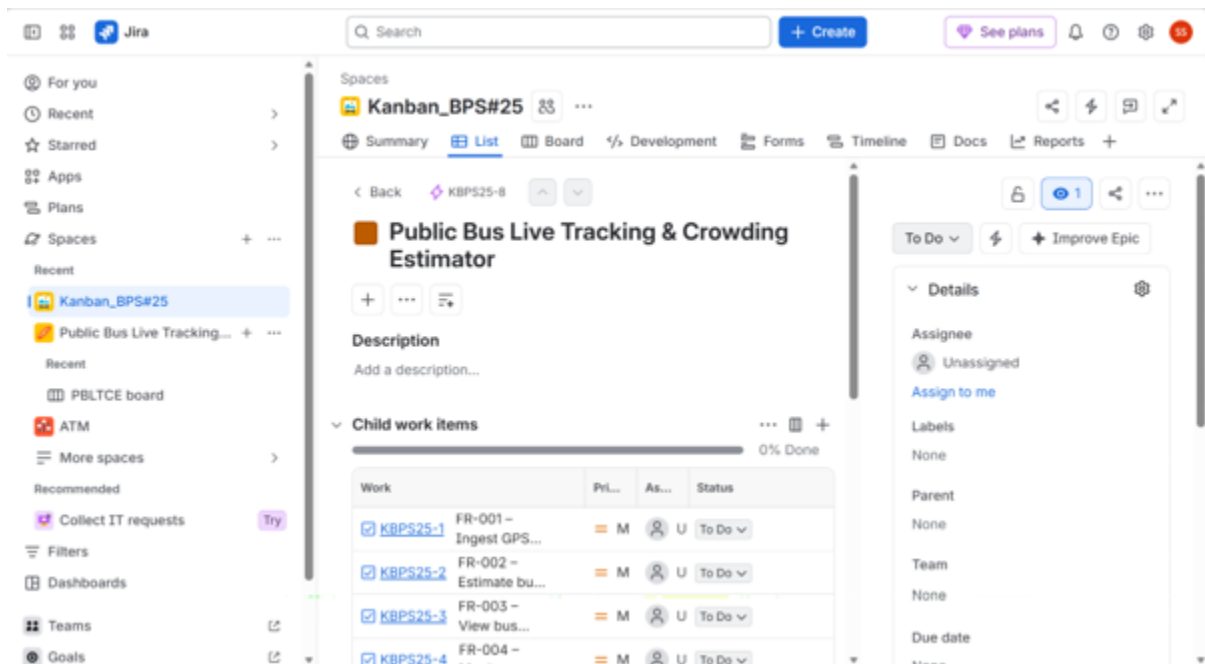
The seven requirement Tasks are:

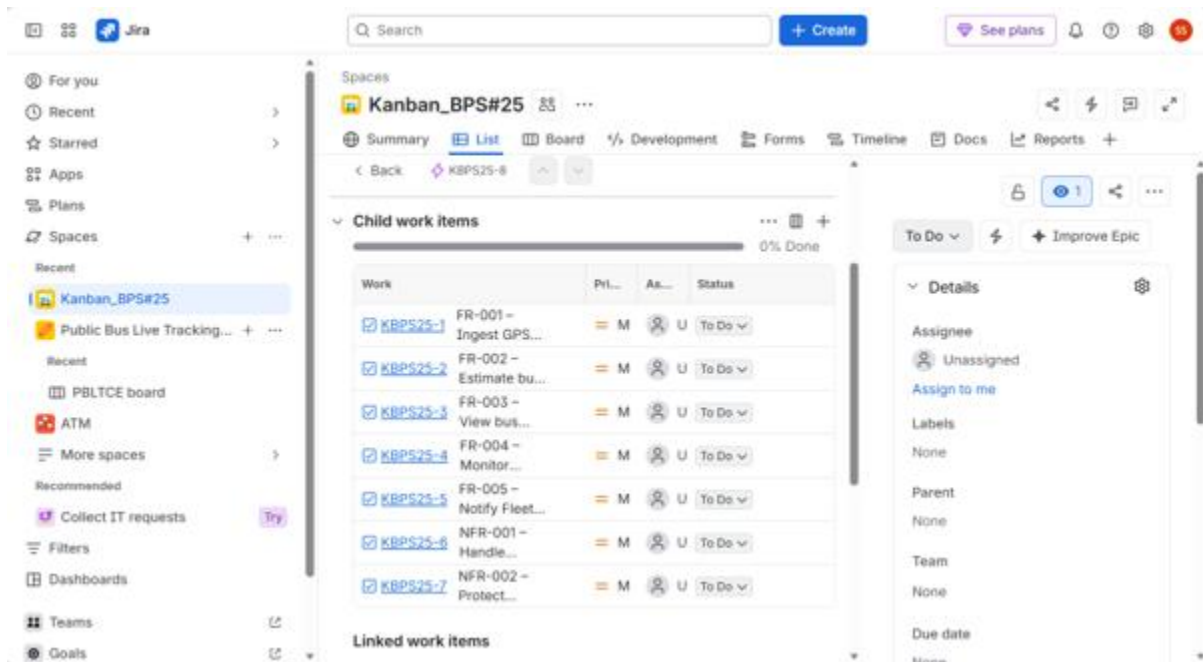
- FR-001
- FR-002
- FR-003
- FR-004
- FR-005
- NFR-001
- NFR-002

Screenshot 2 – Epic and Requirement Tasks

Description:

The screenshot shows the requirement Tasks associated with the **Public Bus Live Tracking & Crowding Estimator** Epic in Jira.





5. STORIES FOR THE REQUIREMENTS

Stories were created for all seven requirements.

Each Story represents the corresponding use case/requirement and was associated with the **Public Bus Live Tracking & Crowding Estimator** Epic.

The Stories created were:

1. **FR-001 – Ingest GPS coordinates from buses**
2. **FR-002 – Estimate bus crowding using ticketing/sensor data**
3. **FR-003 – View bus location, ETA, and crowding information**
4. **FR-004 – Monitor active buses, delays, and high crowding**
5. **FR-005 – Notify Fleet Controller when delay/crowding thresholds are exceeded**
6. **NFR-001 – Handle telemetry from up to 1,000 active vehicles simultaneously**
7. **NFR-002 – Protect GPS/ticketing data and maintain at least 99.5% availability**

Each Story was linked to its corresponding requirement Task using the Jira **relates to** relationship.

Screenshot 3 – Stories Created

Description:

The screenshot shows the Stories created for all five Functional Requirements and two Non-Functional Requirements. The Stories are associated with the project Epic and linked to their corresponding requirement Tasks.

The screenshots show the Jira interface for the 'Kanban_BPS#25' project. The main view is the 'Public Bus Live Tracking & Crowding Estimator' Epic. The left sidebar shows the project structure, including 'Recent' and 'Recommended' items. The right sidebar shows the 'Details' panel for the selected item.

Top Screenshot: Child work items 1-4

Work	FR...	As...	Status
KBPS25-1	FR-001 - Ingest GPS...	M	U To Do
KBPS25-2	FR-002 - Estimate bu...	M	U To Do
KBPS25-3	FR-003 - View bus...	M	U To Do
KBPS25-4	FR-004 - ...	M	U To Do

Bottom Screenshot: Child work items 5-22

Work	FR...	As...	Status
KBPS25-5	FR-005 - Notify File...	M	U To Do
KBPS25-6	NFR-001 - Handle...	M	U To Do
KBPS25-7	NFR-002 - Protect...	M	U To Do
KBPS25-16	FR-001 - Ingest G...	M	U To Do
KBPS25-17	FR-002 - Estimate...	M	U To Do
KBPS25-18	FR-003 - View bus...	M	U To Do
KBPS25-19	FR-004 - Monitor...	M	U To Do
KBPS25-20	FR-005 - Notify...	M	U To Do
KBPS25-21	NFR-001 - Handle...	M	U To Do
KBPS25-22	NFR-002 - Protect...	M	U To Do

6. TASKS AND SUBTASKS

Subtasks were created for all seven requirement Tasks.

Each Subtask was assigned to its corresponding parent Task.

The hierarchy followed in Jira was:

Epic → Requirement Task → Subtask

Functional Requirement Subtasks

- **FR-001 Task**
 - Implement GPS coordinate ingestion
- **FR-002 Task**
 - Implement bus crowding estimation
- **FR-003 Task**
 - Implement commuter bus information display
- **FR-004 Task**
 - Implement fleet bus monitoring
- **FR-005 Task**
 - Implement delay/crowding notifications

Non-Functional Requirement Subtasks

- **NFR-001 Task**
 - Test telemetry capacity for 1,000 vehicles
- **NFR-002 Task**
 - Implement data protection and availability controls

Screenshot 4 – Tasks and Subtasks

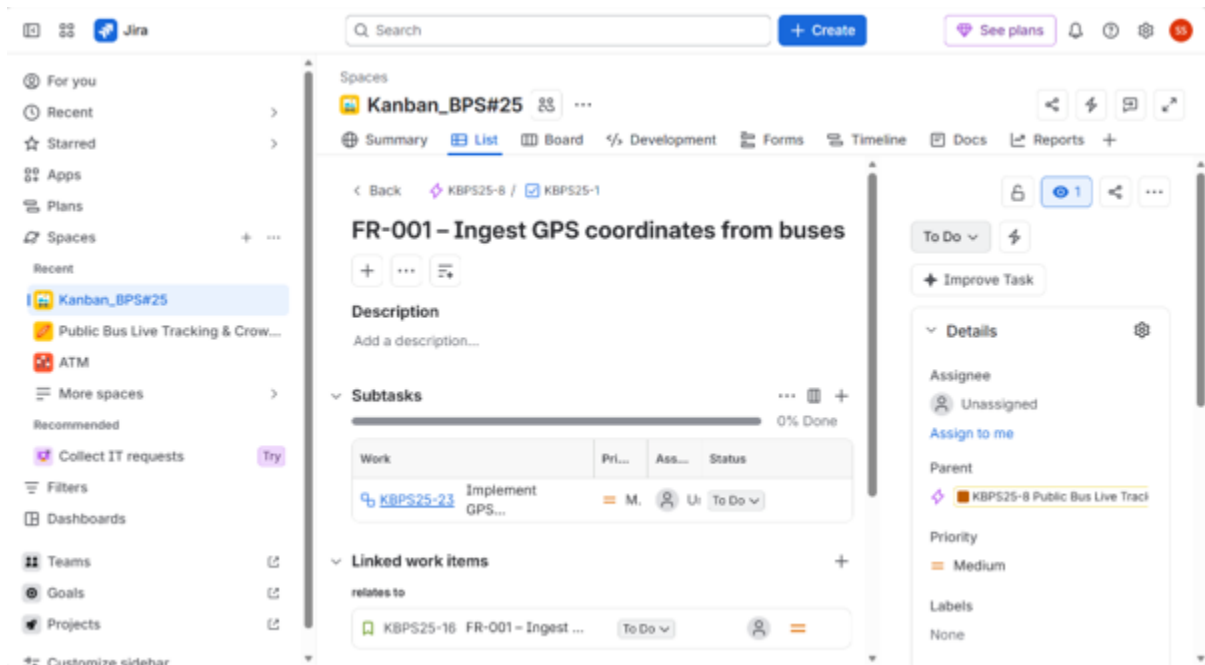
Description:

The screenshots show the requirement Tasks and their corresponding Subtasks in Jira. Each Subtask is associated with the correct parent Task.

Task/Subtask Screenshot Captions

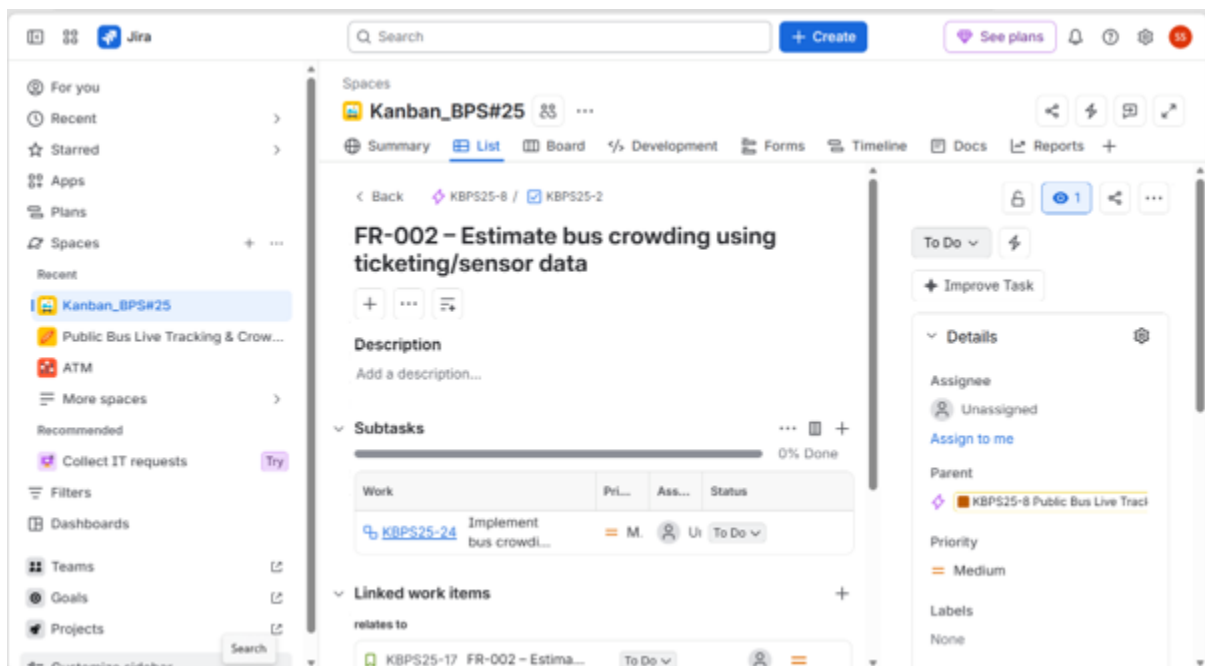
Screenshot 4A – FR-001 Task and Subtask

Shows the FR-001 requirement Task and its corresponding GPS coordinate ingestion Subtask.



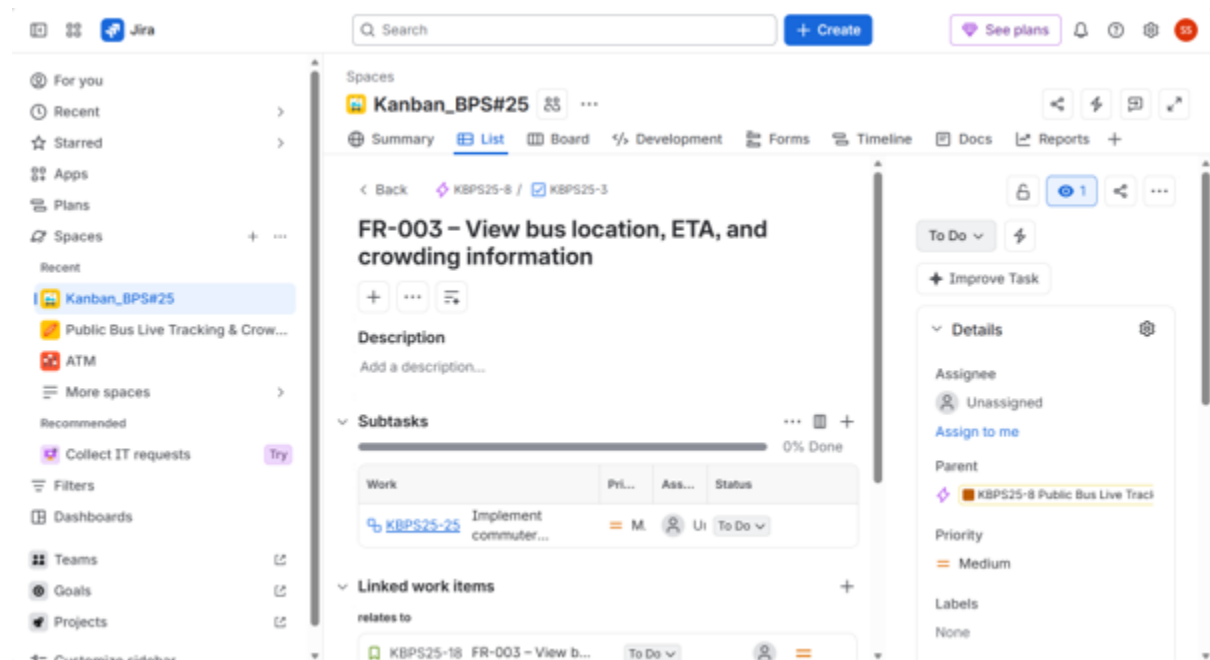
Screenshot 4B – FR-002 Task and Subtask

Shows the FR-002 requirement Task and its corresponding bus crowding estimation Subtask.



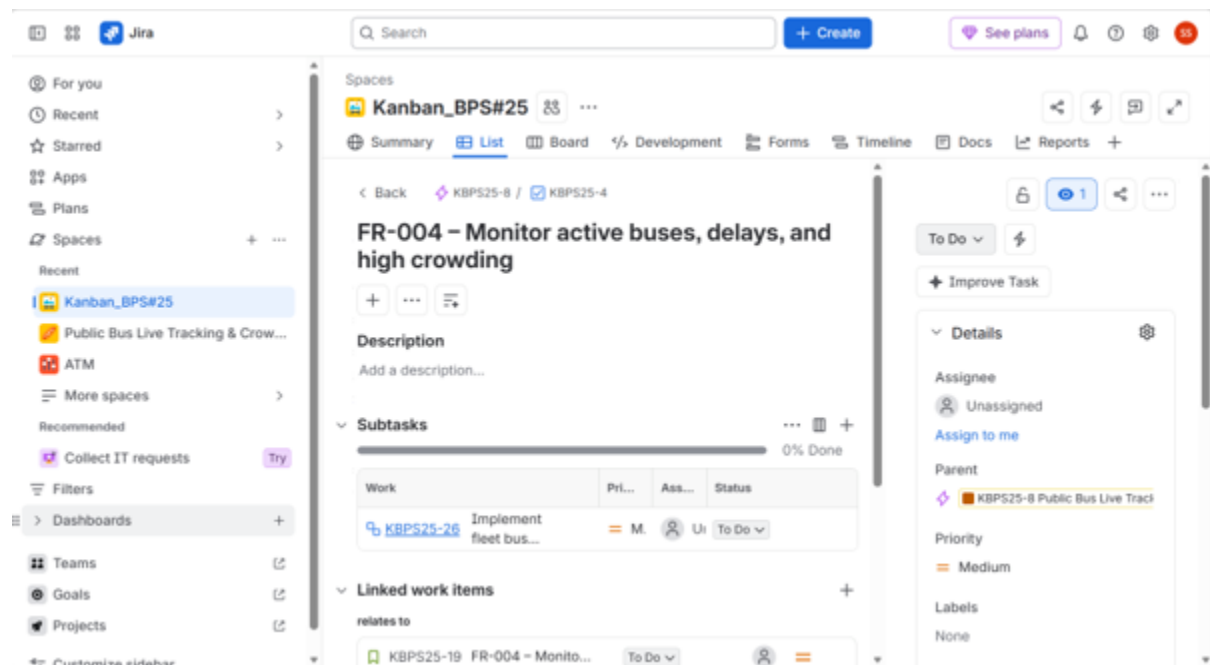
Screenshot 4C – FR-003 Task and Subtask

Shows the FR-003 requirement Task and its corresponding commuter information display Subtask.



Screenshot 4D – FR-004 Task and Subtask

Shows the FR-004 requirement Task and its corresponding fleet monitoring Subtask.



Screenshot 4E – FR-005 Task and Subtask

Shows the FR-005 requirement Task and its corresponding notification Subtask.

The screenshot displays the Jira interface for the 'Kanban_BPS#25' space. The main task is 'FR-005 – Notify Fleet Controller when delay/crowding thresholds are exceeded'. The task details on the right show it is 'To Do', 'Unassigned', and has a 'Medium' priority. The subtasks section shows a progress bar at '0% Done' and a table with one subtask: 'KBPS25-27 Implement delay/crow...'. The linked work items section shows a link to 'KBPS25-20 FR-005 – Notify ...'.

Work	Pr...	Ass...	Status
KBPS25-27 Implement delay/crow...	M	Ui	To Do

Screenshot 4F – NFR-001 Task and Subtask

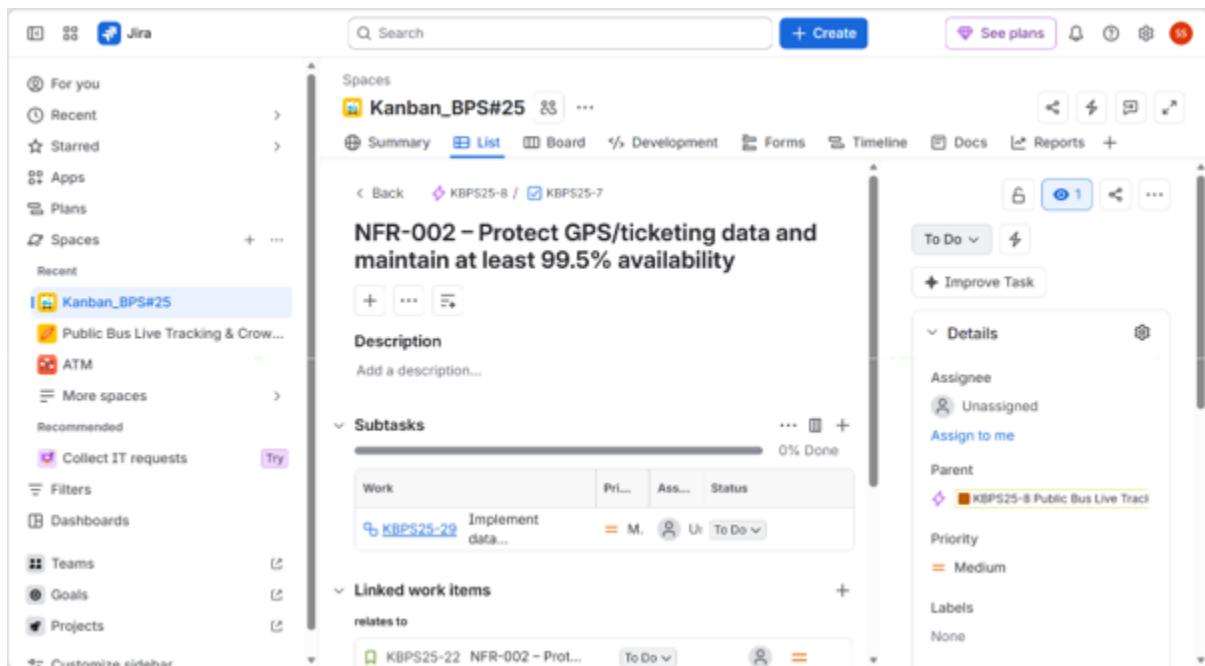
Shows the NFR-001 requirement Task and its corresponding telemetry capacity testing Subtask.

The screenshot displays the Jira interface for the 'Kanban_BPS#25' space. The main task is 'NFR-001 – Handle telemetry from up to 1,000 active vehicles simultaneously'. The task details on the right show it is 'To Do', 'Unassigned', and has a 'Medium' priority. The subtasks section shows a progress bar at '0% Done' and a table with one subtask: 'KBPS25-28 Test telemetry...'. The linked work items section shows a link to 'KBPS25-21 NFR-001 – Handl...'. A tooltip 'To Do: 1 of 1 work items' is visible over the subtask table.

Work	Pr...	Ass...	Status
KBPS25-28 Test telemetry...	M	Ui	To Do

Screenshot 4G – NFR-002 Task and Subtask

Shows the NFR-002 requirement Task and its corresponding data protection and availability Subtask.



7. BACKLOG

The Jira Kanban **Backlog** was used to display the project's work items.

The backlog contains the requirement Tasks and the corresponding Stories. The work items were assigned to the **Public Bus Live Tracking & Crowding Estimator** Epic.

The backlog contains:

- 7 Requirement Tasks
- 7 Stories
- Total: **14 work items**

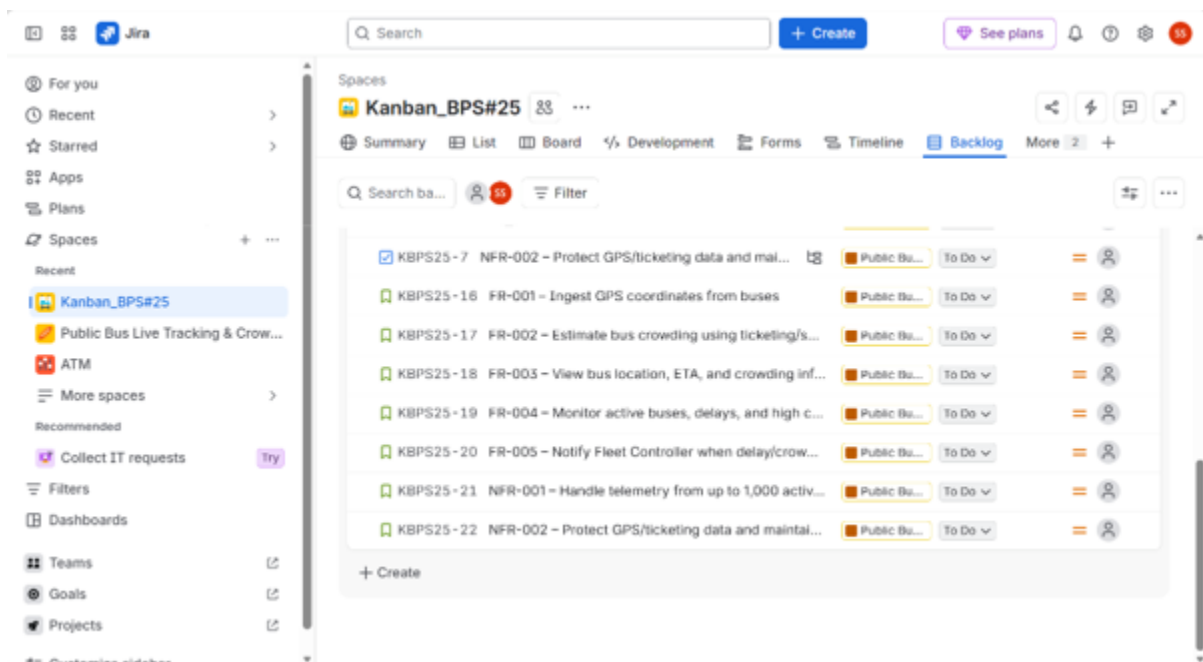
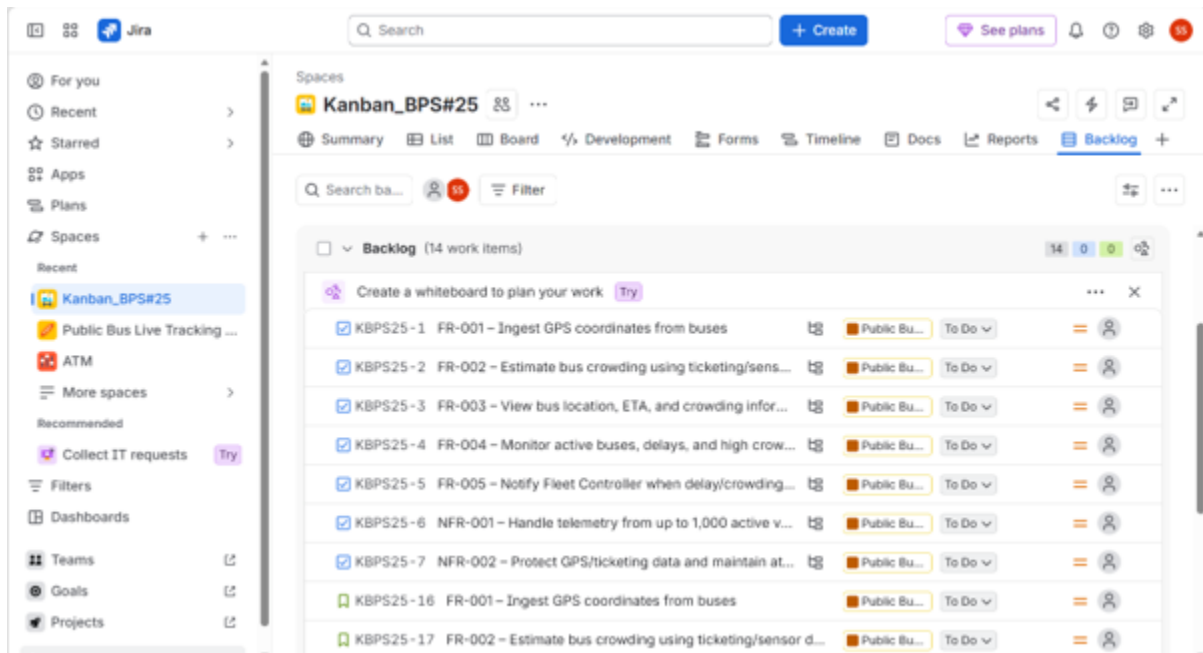
The Epic assignment was verified for the work items.

Screenshot 5 – Kanban Backlog

Description:

The screenshot shows the Kanban backlog containing the requirement Tasks

and Stories. The work items are associated with the **Public Bus Live Tracking & Crowding Estimator** Epic.



8. JIRA WORK ITEM HIERARCHY

The Jira project was organized using the following structure:

Public Bus Live Tracking & Crowding Estimator – Epic



Requirement Tasks

- FR-001
- FR-002
- FR-003
- FR-004
- FR-005
- NFR-001
- NFR-002



Subtasks

Each requirement Task contains its corresponding Subtask.

In addition, each requirement has a corresponding **Story**, which is linked to its requirement Task.

This structure helps organize and track the different requirements and implementation activities in Jira.

9. KANBAN WORKFLOW

The Kanban board provides different stages for tracking the progress of work items.

The main workflow stages used in the project are:

- **To Do**

- **In Progress**
- **In Review**
- **Done**

Work items can be moved between these stages according to their progress.

This helps visualize the current status of requirements and development activities.

10. CONCLUSION

The **Public Bus Live Tracking & Crowding Estimator** problem statement was successfully implemented as a **Kanban project in Jira**.

The project includes:

- 1 Epic
- 5 Functional Requirements
- 2 Non-Functional Requirements
- 7 Requirement Tasks
- 7 Stories
- 7 Subtasks
- Backlog organization
- Task and Story relationships
- Epic assignments

The Kanban board provides a visual representation of the project work and helps in organizing, tracking, and managing the requirements throughout the development process.